Hierarchical Approach for Extracting HPO Codes from Clinical Notes

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Abstract

Rare disease diagnosis poses a significant challenge in clinical settings due to the labor-intensive and error-prone nature of human code assignment, coupled with the limited availability of labeled data which limits the development of accurate predictive models. Existing approaches, including traditional machine learning and deep learning models, often struggle to generalize to rare and underrepresented phenotypes, leading to low accuracy and limited clinical applicability. To tackle this issue, we propose a novel hybrid deep learning model, which we are calling HierTag, that combines hierarchical coding of clinical phenotypes with an abstention mechanism, allowing the model to abstain from uncertain predictions. HierTag is designed to improve classification performance for rare diseases by capturing hierarchical relationships between phenotypic features and enhancing its ability to detect patterns in clinical free text. We evaluate the model on a large-scale dataset of clinical phenotypes and compare its performance to PhenoBERT, an established phenotype classification transformer model. Our results show that the HierTag Model outperforms PhenoBERT in terms of precision and F1-score. Additionally, the abstention mechanism improves the model's reliability in clinical settings. This work presents a promising solution to the data limitations in rare disease diagnosis and sets the stage for future advancements in machine learning-based clinical decision support systems. Future work will focus on expanding the training dataset and refining the model to improve generalizability and robustness for broader clinical applications.

1 Key Information to include

- Mentor: Hee Jung Choi

2 Introduction

Rare diseases affect an estimated 300 million people worldwide [1], yet many patients experience years of misdiagnoses and ineffective treatments before receiving an accurate diagnosis. A key challenge in diagnosing rare diseases is identifying patterns in phenotypic abnormalities, which are deviations from typical physiology, behavior, or morphology. The Human Phenotype Ontology (HPO) provides a standardized framework for representing these phenotypic abnormalities, making it a crucial resource for disease classification, clinical decision support, and biomedical research [2]. However, manually annotating clinical notes with HPO codes is labor-intensive, inconsistent, and prone to errors, leading to variability in diagnoses and downstream analyses. Automating this process is essential for improving diagnostic efficiency, enabling large-scale phenotype-genotype correlation studies, and enhancing the overall reliability of rare disease research.

Despite recent advancements in natural language processing for biomedical text, accurate HPO coding remains challenging due to several factors:

- Hierarchical relationships between phenotypes – Symptoms exist in structured ontologies where general terms ("vision impairment") encompass more specific conditions ("night blindness"). A model must recognize these relationships to avoid overgeneralization or misclassification.
- Context-dependent interpretation – The same term can have different meanings depending on clinical context (ie. "weakness" could refer to muscular, neurological, or systemic conditions). A robust model must infer meaning from surrounding text, rather than relying on simple token matching.

Transformer-based models like PhenoBERT [3] have been applied to automated HPO extraction, but they often fail to capture the hierarchical structure of phenotypic terms, leading to incorrect assignments. Meanwhile, DeepTag [4], a model designed for hierarchical concept tagging, has shown promise in structured classification tasks but has not been fully optimized for phenotype extraction from clinical text.

In this work, we propose a hybrid approach that integrates DeepTag's hierarchical learning capabilities with PhenoBERT's text segmentation strengths. Specifically, we want to develop our own Bidirectional Long Short-Term Memory (BLSTM) network, leveraging DeepTag's structured representation to improve phenotype classification. The key components of this model include:

- A hierarchical training objective, ensuring predictions align with the structured nature of HPO.
- Several modified loss functions that penalize misclassifications based on semantic distance within the ontology.
- An abstention mechanism, allowing the model to defer uncertain predictions rather than producing incorrect classifications.

By integrating hierarchical learning, improved loss functions, and abstention mechanisms, our approach enhances the accuracy, reliability, and interpretability of automated HPO coding, providing a more robust solution for rare disease diagnosis and phenotype-driven research.

3 Related Work

There are various approaches to automating disease coding, with systems like SNOMED-CT and ICD-9/10 commonly used to classify medical conditions. Specifically, the task of automatically extracting HPO codes from clinical text can be addressed using two main methods: dictionary-based approaches and machine learning (ML) or deep learning (DL) techniques.

Dictionary-based methods such as the National Center for Biomedical Ontology annotator [5] rely on predefined lists of terms, such as those found in HPO, to match concepts against the text. These approaches are effective in terms of precision, especially for specific terms within the ontology. However, their major limitation lies in capturing language variations, such as synonyms or alternate expressions of the same condition. As a result, dictionary-based methods often have lower recall since they fail to capture all possible ways a phenotype might be described in clinical notes.

In contrast, ML and DL-based methods have become popular for their ability to understand more complex relationships within the text. One notable example is PhenoBERT [3], developed by Feng et al. (2023), which combines a convolutional neural network for candidate selection with a BERT-based language model for final predictions. By using the CNN module, PhenoBERT reduces the number of potential HPO codes that need to be processed, improving efficiency when working with a large set of over 13,000 possible codes. However, PhenoBERT struggles to understand the context of phenotypes, such as distinguishing between the presence or absence of a condition (ie. "no scoliosis") or determining whether the condition refers to the patient or a family member. These limitations affect the performance of the model in clinical applications in real world.

DeepTag [4], developed by Nie et al. (2018), introduces a hierarchical approach to tagging clinical concepts in free-text notes. DeepTag uses a BLSTM network augmented with a hierarchical training

objective, which allows it to better capture relationships between diagnosis codes. This hierarchical structure is particularly relevant for our work, as it aligns with the hierarchical nature of HPO codes. In addition, DeepTag incorporates an abstention mechanism, allowing the model to abstain from making predictions when uncertain. This feature helps avoid making incorrect or unreliable predictions, which is particularly important in clinical contexts where accuracy is emphasized.

Although DeepTag was designed for veterinary clinical notes, which are typically less complex than human clinical notes, its use of hierarchical learning and the abstention mechanism offers valuable insights for our approach to HPO code extraction. In this work, we aim to combine these features to better address the challenges of phenotype extraction from clinical text.

4 Approach

In this section, we describe our methodology for improving automated HPO coding by integrating hierarchical learning and contextual modeling. We first present our baseline model, PhenoBERT, followed by our proposed hybrid approach model that we've named HierTag, which incorporates a convolutional neural network (CNN)-based encoder, Bidirectional Long Short-Term Memory (BLSTM), ontology-aware loss function, and an abstention mechanism.

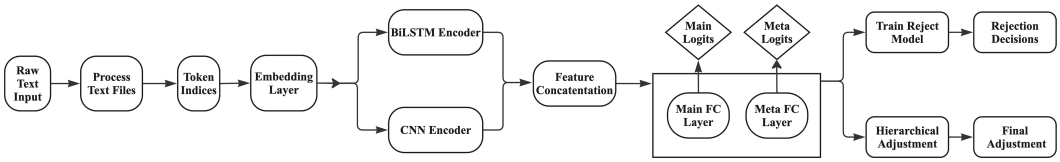


Figure 1: HierTag Model Architecture.

4.1 Baseline: PhenoBERT

PhenoBERT [3] is a transformer-based model designed for automated phenotype extraction from clinical text. We selected it as our baseline because it has demonstrated strong performance in extracting HPO codes. PhenoBERT processes input text in two main stages: text segmentation and a hierarchical CNN-based candidate selection.

In the text segmentation stage, long clinical notes are divided into smaller, manageable chunks known as clinically relevant text segments (CTSs) [6], allowing the model to process information more effectively. Following segmentation, the model applies a bi-layer convolutional neural network (CNN) encoder. The first CNN layer identifies a broad HPO category for each CTS, selecting from one of the 25 child HPO terms under the root node of the HPO ontology. The second level consists of 25 separate CNNs, each corresponding to an HPO sub-graph associated with one of these broad HPO categories.

Once candidate HPO codes are selected, PhenoBERT fine-tunes a BERT encoder, leveraging self-attention mechanisms to capture contextual relationships within the text. This allows the model to determine whether the meaning of a CTS aligns with the description of a given HPO term, improving accuracy in phenotype extraction.

4.2 CNN Encoder for Feature Extraction

In our HierTag Model, we first adopted the same data pre-processing and text segmentation mechanism as PhenoBERT. Then, we introduced our own more advanced CNN-based encoder for text feature extraction. The CNN processes the input tokens using the following operation:

$$h_i = \text{ReLU}(W_{\text{conv}} * x_i + b)$$

where x_i represents the input feature vector at position i, W_{conv} is the convolutional weight matrix, and b is the bias term. The output of the convolutional layer is then subjected to a pooling operation, where we apply the maximum pooling function across the hidden states at each position:

$$x = [\max(h_1), \max(h_2), \max(h_3)]$$

This process extracts key features from the input tokens, making them suitable for downstream processing by the BLSTM and other components.

4.3 Hierarchical Learning with BLSTM

A key part of our approach is hierarchical learning, which we adopted from Nie et al. and their work on DeepTag, and which enables our model to understand and capture dependencies at different levels of the HPO ontology and across the hierarchical structure of the data. We achieve this by integrating a convolutional neural network (CNN) encoder with a Bidirectional Long Short-Term Memory (BLSTM) network. The CNN encoder extracts local features from the input text—capturing broad categories and specific sub-graphs within the HPO—by applying multiple convolutional filters with varying kernel sizes. Simultaneously, the BLSTM processes the sequence of token embeddings in both forward and backward directions:

$$h_f^t = \mathrm{LSTM}_f(x_t,\, h_f^{t-1}) \quad ; \quad h_b^t = \mathrm{LSTM}_b(x_t,\, h_b^{t+1})$$

Instead of simply using the final hidden state, we apply max pooling across all BLSTM outputs to capture long-range dependencies. The resulting BLSTM features are then concatenated with the CNN features to form a unified, rich representation of the input.

Hierarchical learning is further enforced by adjusting the model's raw logits with a precomputed hierarchy matrix. In our model, after obtaining the combined feature representation, the logits are computed via fully connected layers. A hierarchy adjustment is then performed by computing the sigmoid of these logits, multiplying them by the hierarchy matrix to derive parent probabilities, and adding these back to the logits. This mechanism ensures that the predictions respect the hierarchical relationships encoded in the HPO.

In this manner, our model leverages both the CNN's ability to extract local, ontology-relevant features and the BLSTM's capacity for bidirectional contextual modeling, yielding a robust framework for capturing complex dependencies and relationships within phenotype descriptions.

4.4 Loss Functions

To enhance the performance of the model, we added several advanced loss functions that promote more accurate and interpretable predictions:

1. Meta Loss: The meta loss function is used to incorporate meta-learning principles into the training process. It is defined as the binary cross-entropy loss between the predicted and true meta labels:
$$\mathcal{L}_{\text{meta}} = \mathrm{BCE}(p_{\text{meta}}, y_{\text{meta}})$$
This loss promotes learning across different tasks or domains, enhancing the model's generalizability.
2. Co-occurrence Loss: The co-occurrence loss captures the relationships between phenotype terms based on their co-occurrence in the clinical text:
$$\mathcal{L}_{\text{co}} = \sum_{i,j} f(X_{ij})(w_i \cdot w_j + b_i + b_j - \log X_{ij})^2$$
This encourages the model to recognize when certain HPO terms are likely to occur together, improving its understanding of term associations.
3. BCE Loss: Finally, we apply binary cross-entropy (BCE) loss to the final prediction layer:
$$\mathcal{L}_{\text{BCE}} = -\sum_i y_i \log(p_i) + (1 - y_i) \log(1 - p_i)$$
This ensures the model is optimized for accurate classification of HPO terms, balancing precision and recall.

4.5 Abstention Mechanisms

In clinical settings, incorrect diagnoses can have severe consequences, making uncertainty in predictions a critical concern. Inspired by DeepTag, our approach incorporates an abstention mechanism that leverages a reject model to decide when to withhold a prediction. In our implementation, the combined feature representation, obtained by concatenating the outputs of our CNN and BLSTM encoders, is used as input to a reject model, which is a feed-forward neural network. This reject

model is trained to predict the per-example loss (serving as a proxy for prediction confidence) by minimizing the mean squared error between its outputs and the actual per-sample loss computed from the main classifier.

For each example i, the main classifier computes the per-example loss L_i using binary cross-entropy:

$$L_i = \frac{1}{K} \sum_{k=1}^{K} \Big[-y_{i,k} \log\big(p_{i,k}\big) - (1 - y_{i,k}) \log\big(1 - p_{i,k}\big) \Big],$$

where K is the number of labels, $y_{i,k} \in \{0, 1\}$ is the ground truth for sample i and label k, and $p_{i,k} = \sigma\big(\hat{y}_{i,k}\big)$ is the predicted probability obtained by applying the sigmoid function to the logit $\hat{y}_{i,k}$.

The reject model outputs a scalar R_i for each sample and is trained to minimize the following mean squared error:

$$\mathcal{L}_{\text{abst}} = \frac{1}{N} \sum_{i=1}^{N} \Big(R_i - L_i \Big)^2,$$

where N is the total number of samples.

During inference, the trained reject model produces a rejection score for each example. If this score exceeds a predefined threshold, specifically, the mean loss over the validation set, the system abstains from making a prediction for that instance. This approach ensures that when the model is uncertain (as indicated by a high rejection score), the prediction is flagged for human review, thereby reducing the risk of incorrect diagnoses.

4.6 Implementation Details

We implement our model using PyTorch, using the PhenoBERT-associated embeddings that are pre-trained with BioBERT, for the sake of consistency and greater relevancy in a clinical context. Training is conducted using the Adam optimizer with a learning rate of 0.001. We evaluate performance on a dataset of clinical notes annotated with HPO codes, comparing against PhenoBERT using precision, recall, and F1-score metrics.

4.7 Summary

Our approach enhances automated phenotype extraction by combining hierarchical learning, ontology-aware loss functions, and an abstention mechanism. By incorporating advanced CNN-based feature extraction, BLSTM for hierarchical dependencies, and several tailored loss functions, we aim to provide a more accurate and interpretable solution for HPO coding, ultimately improving rare disease diagnosis and research.

5 Experiments

5.1 Data

The datasets utilized in this experiment are sourced from the GitHub repository created by Feng et al., which is associated with the PhenoBERT project. The first dataset, GSC+, consists of 228 biomedical literature abstracts in raw format [7]. The second dataset, ID-68, includes 68 clinical descriptions of real-world cases related to intellectual disabilities [8]. Both datasets have been made publicly available after undergoing rigorous privacy processing to ensure compliance with data protection standards.

These datasets serve both as training and testing resources, providing a diverse range of content that includes both academic (biomedical literature) and clinical (real-world patient descriptions) data. The table below provides an example of input and output data taken from the GSC+ dataset.

In this example, the input consists of a clinical description detailing the patient's medical history, and the output is a set of phenotype terms (in this case, clinical manifestations such as "seizures" or "decrease in fetal movements") alongside their corresponding HPO codes (such as "HP:0001558" for "decrease in fetal movements"). The task involves extracting these HPO codes from the given text, a process which is crucial for automating the phenotypic classification of clinical cases.

Input	Output	
An 8 years old girl who was born at full term via NSVD to G4P2+1 33 years old mother. A decrease in fetal movements was noted during pregnancy, and her birth weight was 2.8 kg. After birth, she was noted to be floppy with poor sucking. At one month of age, she started to have choking episodes that turned to be due to severe GERD, and Niessel fundoplication was performed. Aspiration problems persisted, and the fundoplication procedure was repeated later. She also has developed seizures and continues to seize daily despite multiple attempts to control her seizures with medications.	decrease in fetal movements: with poor sucking: choking episodes: GERD: Aspiration problems: seizures:	HP:0001558 HP:0002033 HP:0030842 HP:0002020 HP:0002835 HP:0001250

Table 1: Example of Input and Output Data from the ID-68 Dataset. The input consists of raw clinical text, and the output includes the corresponding HPO terms and their associated codes.

5.2 Evaluation method

We measure the performance of both our baseline PhenoBERT model and our HierTag Model using precision (P), recall (R), and F1-score (F1). We evaluate the prediction results using both macro-average and micro-average metrics. Macro-average metrics compute the metric independently for each class (such as a clinical note) and then take the average, while micro-average metrics aggregate the contributions of all classes to compute the overall average metric.

5.3 Experimental details

This section details how we ran our experiments, including model configurations, learning rate, and training time.

Model Configuration.

The model uses an embedding dimension of 100, with each word represented by a 100-dimensional vector. The hidden size of the LSTM is 512, meaning the LSTM hidden state has 512 units. The model is built with a single LSTM layer (depth = 1) and is bidirectional, which effectively doubles the output size to 1024. A dropout rate of 0.2 is applied in the CNN encoder and between LSTM layers to prevent overfitting. The CNN kernel sizes are set to [3, 4, 5], allowing the filters to capture various n-gram features. Each kernel generates 100 features, resulting in 100 filters per kernel size. The meta loss weight (β) is set to 0.1, controlling the auxiliary meta loss contribution, while the co-occurrence loss weight (γ) is also 0.1, which scales the penalty for co-occurrence violations. Lastly, the label size is determined by the number of keys in the HPO hierarchy, as specified in the `hpo_hierarchy.json` file.

Training Parameters. The training process involved several key parameters. The learning rate was set to 0.001, used in conjunction with the Adam optimizer. The batch size was set to 32, meaning that mini-batches of 32 samples were processed at each step. The model was trained for a total of 65 epochs, ensuring that it had sufficient time to learn from the data. The overall training time was approximately 28 minutes, and the model was trained on a Google Colab A100 runtime.

5.4 Results

The results from both the GSC+ and ID-68 test sets demonstrate that the HierTag Model consistently outperforms PhenoBERT, particularly in terms of precision and F1-score. In clinical settings, precision is often prioritized over recall to minimize false positives, reducing the risk of misdiagnoses and unnecessary treatments. This is especially critical when incorrect predictions could lead to severe consequences, such as avoidable medical procedures or increased healthcare costs. While recall

remains important for identifying all relevant cases, the need for high precision is particularly relevant in high-stakes clinical applications where incorrect predictions could compromise patient safety.

We present the results of our experiment on the GSC+ and ID-68 test sets in Table 2 and Table 3, respectively.

Method	**Micro Average (%)**			**Macro Average (%)**		
	P	**R**	**F1**	**P**	**R**	**F1**
PhenoBERT	79.25	65.97	72.00	79.11	**70.64**	74.64
HierTag Model	**80.15**	**71.84**	**75.77**	**81.76**	70.05	**75.45**

Table 2: Performance comparison on the GSC+ test set. Note: The bold values indicate the highest values.

Method	**Micro Average (%)**			**Macro Average (%)**		
	P	**R**	**F1**	**P**	**R**	**F1**
PhenoBERT	93.89	**81.57**	87.30	94.09	**79.02**	85.89
HierTag Model	**96.51**	79.94	**87.45**	**96.42**	77.73	**86.07**

Table 3: Performance comparison on the ID-68 test set. Note: The bold values indicate the highest values.

Across both test sets, the HierTag Model consistently achieves higher precision and F1-score compared to PhenoBERT, while PhenoBERT tends to have slightly higher recall. This suggests that PhenoBERT captures a broader range of positive cases but at the cost of increased false positives. In contrast, the HierTag Model is more selective, leading to fewer incorrect classifications and greater overall reliability. The magnitude of improvement in precision, particularly in the ID-68 test set, reinforces the HierTag Model's effectiveness at reducing false positives, making it well-suited for clinical applications where precision is extremely important.

Another interesting pattern is that micro-average performance was generally better than macro-average performance across both models. This suggests that performance is stronger on more frequently occurring classes, while rarer classes are more challenging to classify accurately. The HierTag Model's improvement in micro-averages indicates that it is especially effective in high-support categories, which are often the most clinically relevant. However, the drop in macro-average scores highlights the difficulty of capturing cases underrepresented in the training datasets, suggesting that further refinement may be needed to ensure robustness across all categories.

A key factor contributing to the HierTag Model's high precision is the abstention mechanism, which allows the model to refrain from making predictions when confidence is low. This helps mitigate the risk of incorrect classifications, reducing false positives. However, this explains why our macro-average scores were lower, as the model had less confidence with the "rarer" underrepresented phenotypes and thus chose to abstain. While this trade-off is most probably acceptable in clinical settings where incorrect diagnoses carry significant consequences, a key motivator for automating phenotype coding is to identify these rarer phenotypic abnormalities with more ease and efficiency in order to help with rare disease diagnosis. Though our F1 scores for the HierTag Model were consistently higher than PhenoBERT, meaning that we did not compromise recall too much by using abstention, the abstention mechanism could certainly be adjusted to be more appropriate for rare disease phenotype detection.

Overall, the results exceeded our expectations. While we anticipated the HierTag Model to perform well, we did not expect it to consistently outperform PhenoBERT in both precision and F1-score across the two datasets. These findings suggest that the HierTag Model addresses key limitations of the PhenoBERT model, likely due to its refined approach to the inherently hierarchical coding system. This performance enhancement suggests that the HierTag Model is a robust and promising candidate for clinical applications.

6 Analysis

This section provides a qualitative evaluation of the model, aiming to understand its behavior, successes, and limitations by inspecting key characteristics and outputs.

Ablation Study. To assess the impact of the abstention mechanism, we performed an ablation study by disabling abstention and forcing the model to make predictions in all cases. This led to a significant drop in precision, confirming that abstention plays a critical role in filtering out low-confidence predictions, which would otherwise become false positives. Specifically, precision decreased by an average of 3.4%, while recall slightly increased by 1.7%, as expected. Although higher recall can be beneficial in some cases, the rise in false positives suggests that the abstention mechanism is crucial for maintaining the model's reliability.

Difficult Terms and Misclassifications. The model showed difficulty in distinguishing between certain terms with closely related meanings. For example, it misclassified "muscular hypotonia" as "muscular dystrophy." Both terms involve muscular abnormalities, but with different underlying causes. Another specific error involved confusing "ataxia" (loss of motor coordination) with "dyskinesia" (involuntary movements), highlighting the model's struggle to differentiate between nuanced but related movement disorders. These issues suggest that while the model captures broad category relationships well, the hierarchical structure still needs refinement to distinguish clinically similar terms more effectively.

Moreover, the model faced challenges with rare and underrepresented terms, such as "pontocerebellar hypoplasia," likely due to its infrequent appearance in the training data. To improve generalizability, additional targeted training on rare phenotypic abnormalities is needed.

Performance on Different Phenotype Categories. When examining performance across different phenotype categories, we observed that the HierTag Model performed well in high-support categories (like neurological and metabolic disorders) with abundant training data. However, its performance was weaker for rare syndromic phenotypes, which aligns with the observed drop in macro-average performance discussed in the results. This highlights the need for additional training data or specialized fine-tuning, particularly for rare phenotype classes, to improve the model's generalizability and ensure robust performance across all categories.

7 Conclusion

This study aimed to address the challenge of accurate phenotype classification in clinical settings, particularly for rare and complex diseases, where conventional models often struggle. By introducing the HierTag Model for phenotypic coding, we demonstrated that it significantly outperforms the PhenoBERT model in precision and F1-score, especially in contexts where accuracy is critical for diagnosis and treatment planning.

The improved performance of the HierTag Model underscores the importance of hierarchical coding and a refined approach to phenotype categorization, as emphasized in the introduction. The model's integration of an abstention mechanism further strengthens its potential for real-world clinical applications, minimizing false positives while ensuring confidence in predictions. However, challenges remain with rare and underrepresented phenotypes, an issue we plan to address in future work by expanding training datasets to improve generalizability.

These findings contribute to the ongoing efforts to improve AI-driven diagnostic tools, reinforcing the importance of developing models tailored to the specific needs of rare disease diagnosis and precision medicine. Moving forward, additional enhancements to model robustness and training methodology will be key to achieving broader applicability in clinical practice.

Team contributions

Brooke Ballhaus developed the `DeepTag_Model.py` file, which includes the BLSTM and CNN models, hierarchical learning, abstention, configurations, and the training class. She wrote the Approach and Analysis sections of the paper.

Riya Narain developed the `Setup.py` and `HPO_Processing.py` files, which include the creation of the HPO code hierarchy and all data preprocessing. She also developed the evaluation class for the model and wrote the Introduction, Related Work, Experiments, and Conclusion sections of the paper.

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